Diffusion Estimation and MSD/VACF Consistency Specification

G2/GK3 after H0: Explicit Plateau Selection and Signature-Checked Comparison

mdstats

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Contents

Purpose and status

This document specifies the H0-hardened functions in

```
mdstats/analysis/diffusion.py
```

```
estimate_diffusion_plateau(...)
compare_msd_vacf_diffusion(...)
```

and the immutable `DiffusionEstimate` and `DiffusionComparisonResult` schemas. The module consumes existing MSD and running Green-Kubo results. It does not recompute trajectories, correlations, or transport integrals.

Borrowed theory and numerical methods

For a physical subspace of rank d, the Einstein relation gives [1]

$$M_B(t) \sim 2dD_Bt.$$

Thus a linear slope m_B yields

$$D_B = \frac{m_B}{2d}.$$

The compared VACF estimate follows Green [2] and Kubo [3]. The centered ordinary-least-squares formulas used for descriptive slope diagnostics are standard regression algebra. Explicit interval selection, refusal to infer an uncertainty from adjacent running-integral samples, semantic-signature checking, and the symmetric relative difference are mdstats policies.

Explicit plateau estimate

Public function

```
def estimate_diffusion_plateau(
    running: VACFDiffusionResult,
    *,
    time_range_ps: tuple[float, float] | None = None,
    minimum_points: int = 8,
    slope_tolerance: float | None = None,
    method: Literal["explicit", "stable_window"] = "explicit",
) -> DiffusionEstimate:
    ...
```

Only `method="explicit"` is implemented. `"stable_window"` raises `NotImplementedError` until a separately specified automatic-window stage.

The requested interval is inclusive within floating-point boundary tolerance and must lie inside stored lag times. At least `minimum_points` selected samples are required.

Uniform-grid requirement

The current estimator is the arithmetic mean

$$\hat{D} = \frac{1}{N} \sum_{j=1}^{N} D(t_j).$$

H0 requires the selected lag samples to be uniformly spaced. If their increments are not equal within strict tolerance, the function raises. It does not allow a dense region of an irregular grid to receive more implicit weight. A future time-weighted interval estimator must use another method name.

Diagnostics

The function stores descriptive values over the selected interval:

- mean, median, sample standard deviation, minimum, maximum, and span;
- endpoint drift;
- centered linear slope and intercept;
- R^2 and residual RMS;
- selected sample spacing; and
- an optional pass/fail result for the absolute slope tolerance.

`standard_error_a2_per_ps` remains `None`. Adjacent values on one cumulative curve are serially correlated and are not treated as independent replicas.

Diffusion estimate result

```
@dataclass(frozen=True, slots=True)
class DiffusionEstimate:
    value_a2_per_ps: float
    standard_error_a2_per_ps: float | None
    time_range_ps: tuple[float, float]
    method: str
    component: str
    dimensions: int
    n_points: int
    is_stable: bool | None
    diagnostics: Mapping[str, Any]
    metadata: Mapping[str, Any]
    projection_basis: NDArray[np.float64]
    projection_labels: tuple[str, ...] | None
    signature: DynamicsInputSignature | None
```

The basis rank equals `dimensions`. The component label and optional signature must agree with that subspace. Diagnostics and metadata are recursively immutable. The cm^2/s property uses the exact package conversion factor 10^{-4}.

MSD/VACF comparison

Public function

```
def compare_msd_vacf_diffusion(
    msd: MSDResult,
    vacf_diffusion: DiffusionEstimate,
    *,
    msd_fit_range_ps: tuple[float, float],
    dimensions: Literal[1, 2, 3] | None = None,
) -> DiffusionComparisonResult:
    ...
```

The optional `dimensions` argument is deprecated and acts only as a consistency check against the stored subspace rank.

Admission contract

The comparison requires:

- `msd.mode == "time_averaged"`;
- `msd.coordinate_mode == "laboratory"`;
- complete `DynamicsInputSignature` objects on both inputs; and
- equality of every semantic field after assigning the VACF estimate subspace to the MSD signature.

This detects mismatches in source identity, exact frame sequence and times, measured atoms, coordinate/reference-cell treatment, drift mode, exact drift- reference atoms, velocity source, and physical projection.

Unsigned legacy results fail closed.

Projected MSD fit

The selected VACF estimate basis is also used for the MSD:

$$M_B(t) = \mathrm{tr}[BM(t)B^T].$$

Canonical axis subsets may use stored components. A rotated basis requires the full MSD tensor. At least three stored MSD points are required. Centered OLS with an intercept produces slope m_B, then

$$D_{\mathrm{MSD}} = \frac{m_B}{2d}.$$

Difference measures

The result stores

$$\Delta D = D_{\mathrm{MSD}} - D_{\mathrm{VACF}},$$

$$|\Delta D|,$$

and the symmetric relative difference

$$\delta_{\mathrm{sym}} = \begin{cases} 0, & |D_{\mathrm{MSD}}| + |D_{\mathrm{VACF}}| = 0, \\ \frac{2|D_{\mathrm{MSD}} - D_{\mathrm{VACF}}|}{|D_{\mathrm{MSD}}| + |D_{\mathrm{VACF}}|}, & \text{otherwise} \end{cases}.$$

Neither estimator is declared authoritative.

Comparison result

```
@dataclass(frozen=True, slots=True)
class DiffusionComparisonResult:
    msd_diffusion_a2_per_ps: float
    vacf_diffusion_a2_per_ps: float
```

```
    signed_difference_a2_per_ps: float
    absolute_difference_a2_per_ps: float
    symmetric_relative_difference: float
    msd_fit_range_ps: tuple[float, float]
    msd_slope_a2_per_ps: float
    msd_intercept_a2: float
    component: str
    dimensions: int
    n_msd_points: int
    diagnostics: Mapping[str, Any]
    metadata: Mapping[str, Any]
    projection_basis: NDArray[np.float64]
    projection_labels: tuple[str, ...] | None
    signature: DynamicsInputSignature
```

The constructor validates all algebraic difference identities, subspace fields, fit interval, point count, and signature projection. Arrays and nested mappings are immutable.

Interpretation policy

The module reports, but does not automatically resolve:

- nonpositive MSD slopes;
- a VACF interval that fails the requested slope tolerance;
- a VACF interval whose stability was not assessed;
- imperfect linearity in the MSD fit; and
- finite-record differences between Einstein and Green-Kubo estimates.

No automatic ballistic/caged/diffusive classification is performed.

Required tests

- constant and gently drifting running curves produce expected diagnostics;
- irregular selected lag grids raise;
- boolean point counts and invalid tolerances raise;
- an unsupported automatic window raises explicitly;
- scalar 3D and Cartesian comparison values preserve pre-H0 references;
- axis-subset and rotated MSD projections use the stored VACF subspace;
- a rotated projection without the MSD tensor raises;
- different drift-reference atoms raise;
- different slices under the same filename raise;
- coordinate/reference-cell and velocity-source mismatches raise;
- unsigned results raise;
- the deprecated `dimensions` check cannot reinterpret the subspace;
- difference identities and units are exact; and
- arrays, diagnostics, and metadata are deeply immutable.

References

[1] A. Einstein, *Ann. Phys.* **322**, 549-560 (1905). DOI: 10.1002/andp.19053220806.

[2] M. S. Green, *J. Chem. Phys.* **22**, 398-413 (1954). DOI: 10.1063/1.1740082.

[3] R. Kubo, *J. Phys. Soc. Jpn.* **12**, 570-586 (1957). DOI: 10.1143/JPSJ.12.570.